DBT PORTFOLIO PROJECT

LAYER: INTERMEDIATE

DBT Intermediate Model

GARCH | Volatility | Risk Analytics | Time Series | Jinja2 | Statistics

Project Summary

Approximates a GARCH(1,1) (Generalized Autoregressive Conditional Heteroskedasticity) model in pure SQL to estimate time-varying volatility in healthcare financial returns. Seeds unconditional variance using the omega/(1-alpha-beta) formula, then applies the GARCH(1,1) recursion via LAG(). Outputs conditional variance and volatility per entity and period for risk dashboards.

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1 PROJECT OVERVIEW

Model Name:	int_garch
Layer:	Intermediate
Materialization:	view
GARCH Parameters:	omega=0.000001, alpha=0.09, beta=0.90 (configurable via Jinja vars)
Key Outputs:	conditional_variance, conditional_volatility per entity/date
Audience:	Actuaries, Risk Analysts, Finance leadership

2 BUSINESS PROBLEM

Healthcare financial returns (claim ratios, payer margins) exhibit volatility clustering -- calm periods are followed by calm, turbulent periods by turbulent. Standard deviation assumes constant volatility, which misrepresents risk. GARCH(1,1) captures this time-varying volatility, giving actuarial teams a more accurate picture of when financial risk is elevated and when it is low.

- Volatility clustering: GARCH models the fact that large shocks are followed by large shocks.
- Conditional variance: sigma^2_t changes each period based on recent squared returns.
- Risk dashboards: conditional_volatility plotted over time flags stress periods for management.
- Jinja2 parameters: omega, alpha, beta set at the top of the file -- no SQL edits needed to retune.
- NULLIF guard: prevents division by zero when (1 - alpha - beta) = 0 (integrated GARCH case).

3 TECHNICAL EXPLANATION

- returns CTE: Computes eps_sq = return_value^2 (squared innovations) and assigns row numbers.
- unconditional_var CTE: Seeds sigma^2_0 = omega / (1 - alpha - beta) -- the long-run average variance.
- base CTE: Joins the seed variance onto every row of the return series.
- garch_approx CTE: Applies the GARCH(1,1) update: sigma^2_t = omega + alpha*eps^2_{t-1} + beta*sigma^2_{t-1} using LAG() for both terms.
- SQL approximation note: True GARCH requires recursive computation. LAG() uses observed eps^2 rather than recursively computed sigma^2 -- sufficient for exploratory analytics and feature engineering.
- Final SELECT: COALESCE(sigma_sq_approx, sigma_sq_0) handles the first row with no lag. SQRT() converts variance to volatility.

Key Techniques: Jinja2 set variables, unconditional variance seed, LAG() GARCH recursion, COALESCE null-handling, SQRT() volatility

4 FULL MODEL CODE

```
{{ config(materialized='view') }}

{% set omega = 0.000001 %}
{% set alpha = 0.09 %}
{% set beta  = 0.90 %}

with returns as (
    select
        entity_id,
        period_date,
        return_value,
        power(return_value, 2) as eps_sq,
        row_number() over (partition by entity_id order by period_date) as rn
    from {{ source('healthcare', 'claims') }}
),

unconditional_var as (
    select
        entity_id,
        {{ omega }} / nullif(1 - {{ alpha }} - {{ beta }}, 0) as sigma_sq_0
    from returns
    group by entity_id
),

base as (
    select r.*, u.sigma_sq_0
    from returns r
    join unconditional_var u using (entity_id)
),

garch_approx as (
    select
        entity_id,
        period_date,
        return_value,
        eps_sq,
        sigma_sq_0,
        {{ omega }}
            + {{ alpha }} * lag(eps_sq,    1) over (partition by entity_id order by period_date)
            + {{ beta  }} * lag(sigma_sq_0,1) over (partition by entity_id order by period_date)
            as sigma_sq_approx
    from base
)

select
    entity_id,
    period_date,
    return_value,
    eps_sq,
    coalesce(sigma_sq_approx, sigma_sq_0) as conditional_variance,
    sqrt(coalesce(sigma_sq_approx, sigma_sq_0)) as conditional_volatility
from garch_approx
order by entity_id, period_date
```

entity_id	period_date	return_value	eps_sq	cond_variance	cond_volatility
E-001	2025-01	0.032	0.001024	0.000010	0.003162
E-001	2025-02	0.041	0.001681	0.000010	0.003163
E-001	2025-03	-0.018	0.000324	0.000010	0.003163
E-002	2025-01	0.055	0.003025	0.000010	0.003162

** GARCH parameters omega=0.000001, alpha=0.09, beta=0.90 applied. Tune via Jinja set variables at model top.*